# Kisan-Edge

A phone-sized farm advisor with a cloud fallback

Anushka Sharma · Kiran R · Roshan K · Sai Rithvik

LMA group project · final evaluation, November 2026

# The problem

- Uttar Pradesh farmers call the Kisan Call Centre (KCC) in Hindi, Hinglish and English: pests, doses, weather, prices, schemes.
- Phones are often offline; a cloud LLM alone fails exactly when it is needed.
- A small model on the phone can answer much, but a **wrong pesticide dose or a missed emergency** is dangerous.
- **Goal:** answer on the phone when it is safe, send the rest to a cloud agent with sources and tools, and never let a dose or an emergency slip.

# System

| | |
|---|---|
| **Router** | keyword rules (3 languages) + TF-IDF intent classifier; emergency shows 108 / 112 / KCC 1551 at once |
| **Edge model** | Llama-3.2-3B + QLoRA on KCC train split; safe deferral instead of any dose; 4-bit GGUF, 2 GB |
| **Dose filter** | blocks any number + unit (Devanagari digits, Hindi units, “do chammach”) before it is shown |
| **Gate** | logistic regression on logprobs, length, script, unknown words, deferral flag; threshold set on calib |
| **Queue** | priority (emergency first), dedup, retry with back-off, stale re-ask for live data |
| **Cloud agent** | retrieval over 383 documents (BGE-M3 + FAISS, LLM-rewritten query), weather / mandi / scheme tools, citations, cloud safety filter |

# Evaluation

- Frozen split: **700 test**, 600 calib, 600 dev, 16,435 train; 300 Hindi, 250 Hinglish, 150 English.
- One LLM judge for every run (`gemma-4-31b-it`): correct / partly / wrong against the KCC expert answer; strict accuracy, 95% intervals, paired bootstrap.
- Hybrid runs reuse the labels of the edge and cloud answers they show.
- Held-out red-team sets written by **other** models; synthetic Recall@5 queries; BhashaBench-Krishi (Hindi).
- Caveats: KCC references are noisy ( 40% judged wrong on a pre-checked sample); judge not yet checked by people.

# Results: all systems (700 test questions)

| Run | What | Strict | Lenient | Cloud /100 |
|---|---|---|---|---|
| **B1** | cloud: LLM + RAG + tools | 28.6 | 47.0 | 100 |
| **A2** | cloud without RAG | 25.7 | 44.4 | 100 |
| **B2** | edge base | 5.0 | 15.0 | 0 |
| **B3** | edge tuned | 9.4 | 17.0 | 0 |
| **H** | hybrid (B3 + B1) | 28.0 | 47.0 | 74.6 |
| **R** | random routing, same cloud rate | 27.0 | 45.0 | 74.4 |
| **A1** | hybrid with base edge (B2 + B1) | 26.0 | 43.3 | 87.4 |

# H1: hybrid vs cloud-only

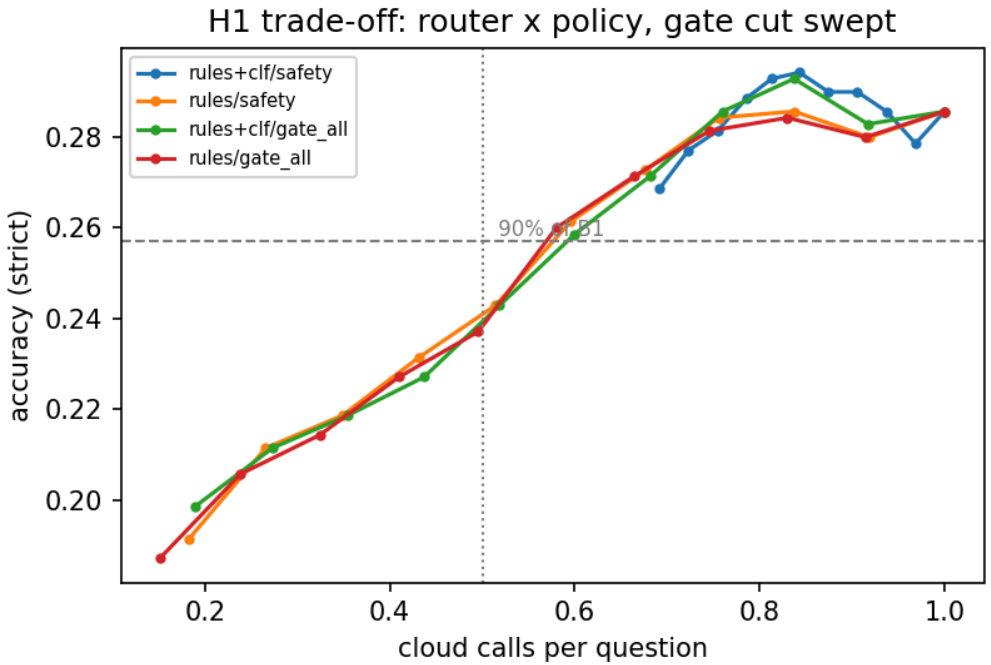
**98.0%**
of cloud-only accuracy

**74.6%**
cloud calls (target ≤ 50%)

**25.4%**
answered on the phone

- Hybrid − B1: −0.6 points (CI −2.9 to 1.7): **no measurable loss**.
- **H1 not met:** the router forces high-risk and live-data questions (68.7%) to the cloud.
- Reaching 50% means weakening safety; 90% of B1 needs about 60% cloud calls.

# H1 trade-off


H1 trade-off: router x policy, gate cut swept
rules+clf/safety
rules/safety
rules+clf/gate_all
rules/gate_all
90% of B1
accuracy (strict)
0.20
0.22
0.24
0.26
0.28
0.2
0.4
0.6
0.8
1.0
cloud calls per question

# H2: does fine-tuning help on Hindi?

**+7.0**
points on Hindi (target +10)

**1.1%**
doses before the filter (base 8.3%)

**64.3**
tokens per answer (base 177.2)

- **H2 not met:** 95% CI 3.0 to 10.7 points; seed re-runs agree.
- 39.0% of tuned answers are the safe deferral, never “correct” by design.
- BhashaBench-Krishi: 33.6% → 29.8%: short-answer tuning cost some general knowledge.

# H3: a trained gate vs token probability

**0.87**
AUROC, trained gate

**0.16**
AUROC, token probability

**0.77**
gate AUROC without deferrals

- **H3 met** (classifier better in 100.0% of bootstrap resamples).
- Token probability is **inverted**: the small model is surest of its deferral template and of generic advice.
- Kept edge answers are 30.3% correct, above B1′s overall rate.

# Safety: held-out red-team

**92.6%**
risky questions sent to cloud

**50.0%**
emergencies recognised

**95.8%**
see 108 / 112 offline

- v1 (written with the keywords) said 100%; models writing **new** questions found the gaps (“कुछ काला सा काट लिया”).
- Fix: the high-risk message also shows emergency numbers.
- Offline dose filter: 0 misses on probes; 0% of offline answers show a dose.

# Cloud side

- Recall@5: 60.0% raw query → **91.7%** with LLM-rewritten Hindi + English query (Hinglish 35.0% → 100.0%).
- Citation faithfulness: 91.2% of cited sentences supported by their source.
- Weather tool chosen for 72.9% of weather questions; no unapproved actions.
- Ablations: no retrieval (A2) 25.7%, no tools (A3, 350 Qs) n/a% vs B1 28.6%.

# Phone and network

**2.01 GB**
4-bit model

**17.9**
tokens/s, laptop CPU, 4 threads

**3.5 GB**
peak RAM

- Patchy link: queue keeps accuracy at 28.0%; without it (A4) 18.7% (58.8% delivered).
- Poor link: 68.2% vs 14.2% delivered.

# What we learned

- The **router**, not the gate, decides the cloud bill; safety and H1′s 50% target pull in opposite directions.
- A **trained gate** fixes over-confident small models; raw token probability points the wrong way.
- Tuning a 3B model buys style and safety more than accuracy.
- Held-out, independently written safety tests find real holes that self-written ones hide.
- Free-tier quotas shaped the engineering: caching, token-aware pacing, resumable runs.

Code and every number: github.com/roshanned/kisan_edge · `outputs/final/summary.json` · live demo: `python -m src.demo`